

Why GAN

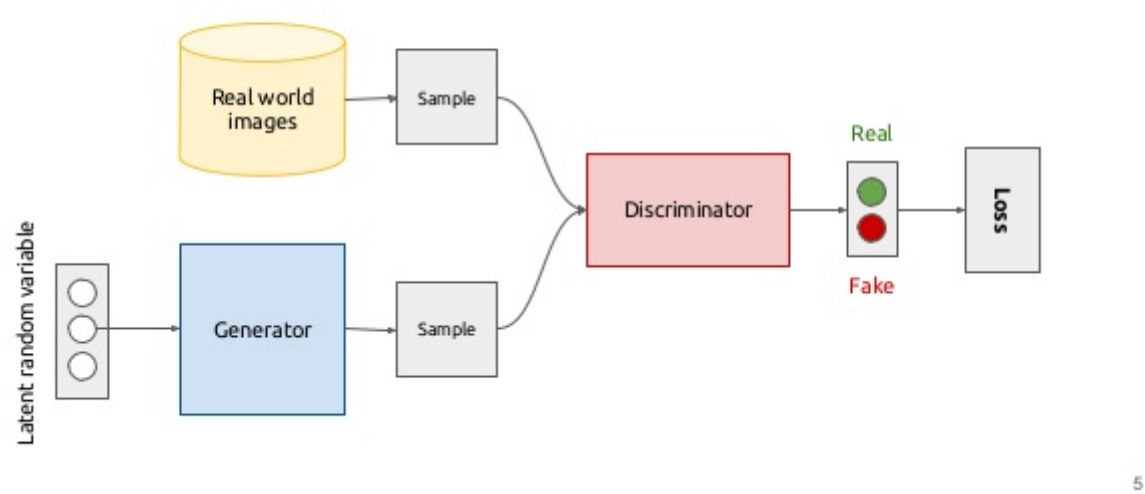
- Only involve neural network and back propagation.
- Generate much more realistic and diverse pictures.
- Can control feature(Z) to generate specific data.
- Promising in the field of unsupervised learning.

What is GAN

Generative adversarial networks are a branch of **unsupervised machine learning**, implemented by a system of **two neural networks competing against each other** in a zero-sum game framework. They were first introduced by **Ian Goodfellow et al.** in 2014.

basic structure

Generative adversarial networks (conceptual)



Generator and Discriminator

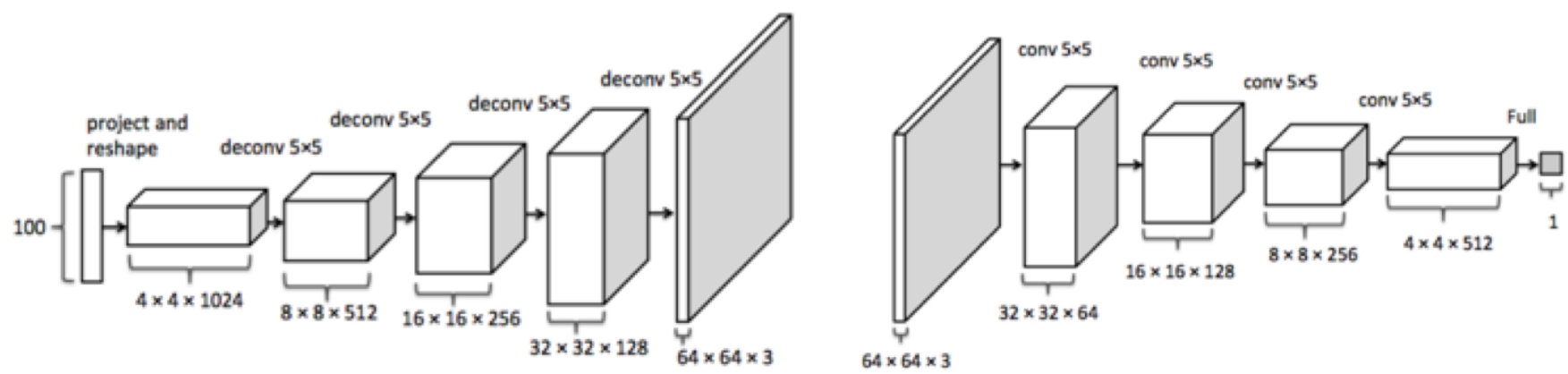


Figure: "Semantic Image Inpainting with Perceptual and Contextual Losses"

Brief Theory

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim P_{data}(x)} [\log D(x)] + \mathbb{E}_{z \sim P_z(z)} [\log(1 - D(G(z)))]$$

- If $P_{data}(x) > P_G(x)$, then x is a point with higher probability of coming from the data than being a generated sample.
- If $P_{data}(x) < P_G(x)$, then x has low probability of being a data point, but high probability of being generated by our model.

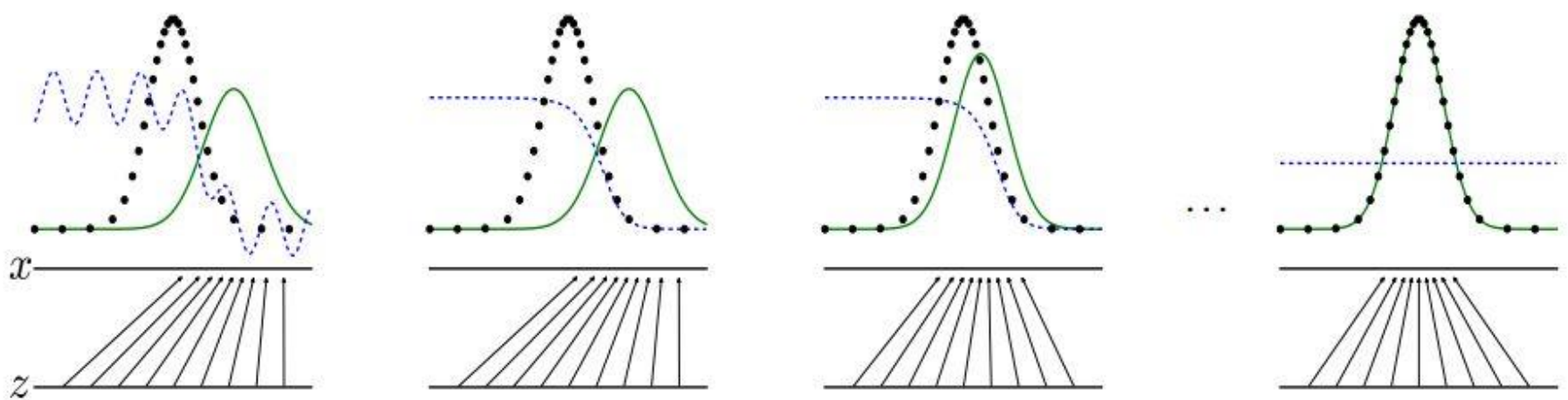


Figure: "Generative Adversarial Nets"

Training Result

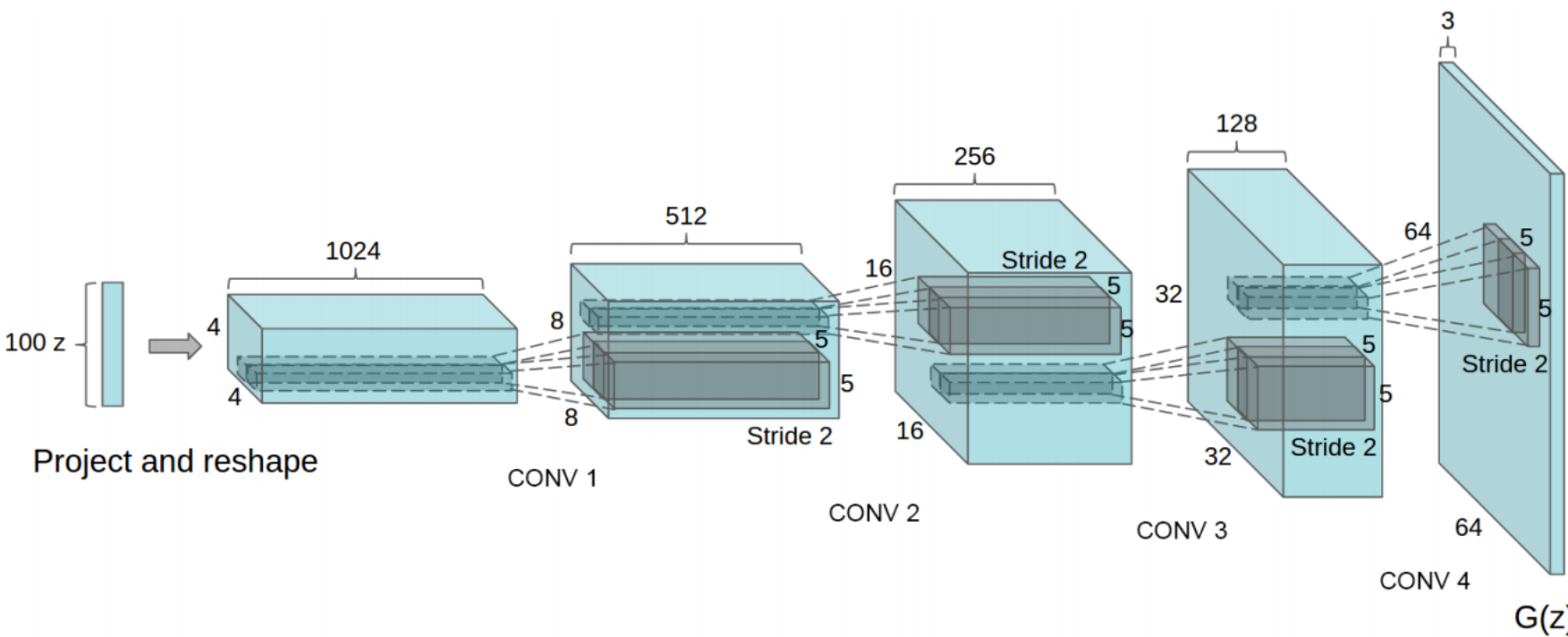
- Generator: $P_G(x) = P_{data}(x)$
- Discriminator: $D(x) = \frac{P_{data}(x)}{P_{data}(x)+P_G(x)}$

Detail Proof: "Generative Adversarial Nets"

Applications

Unclassified

DCGAN



Vector arithmetic on the z input space

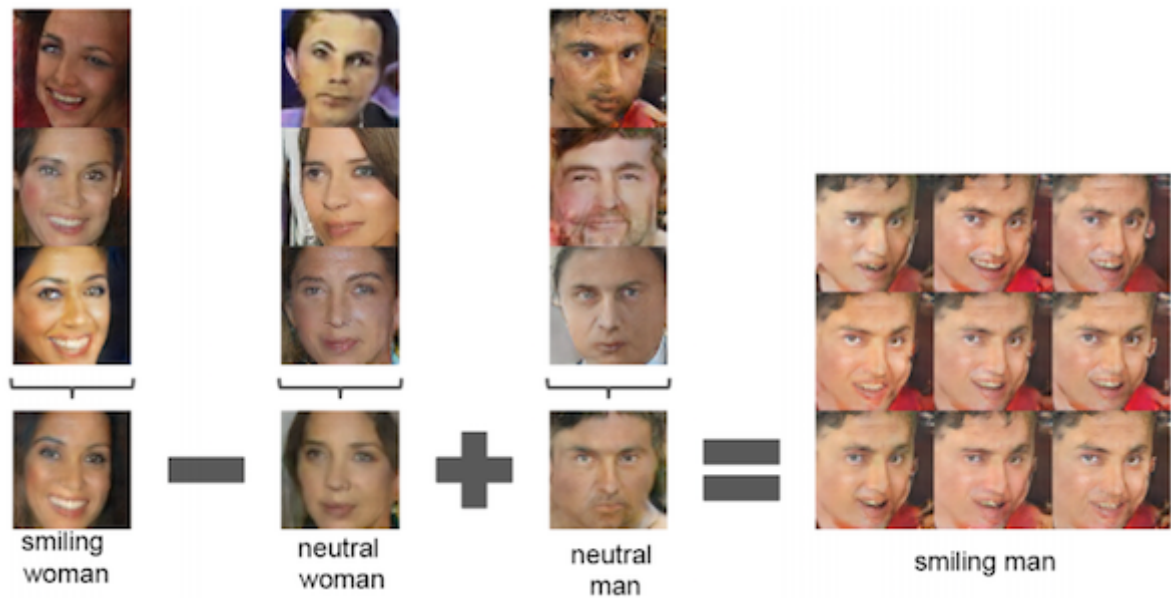
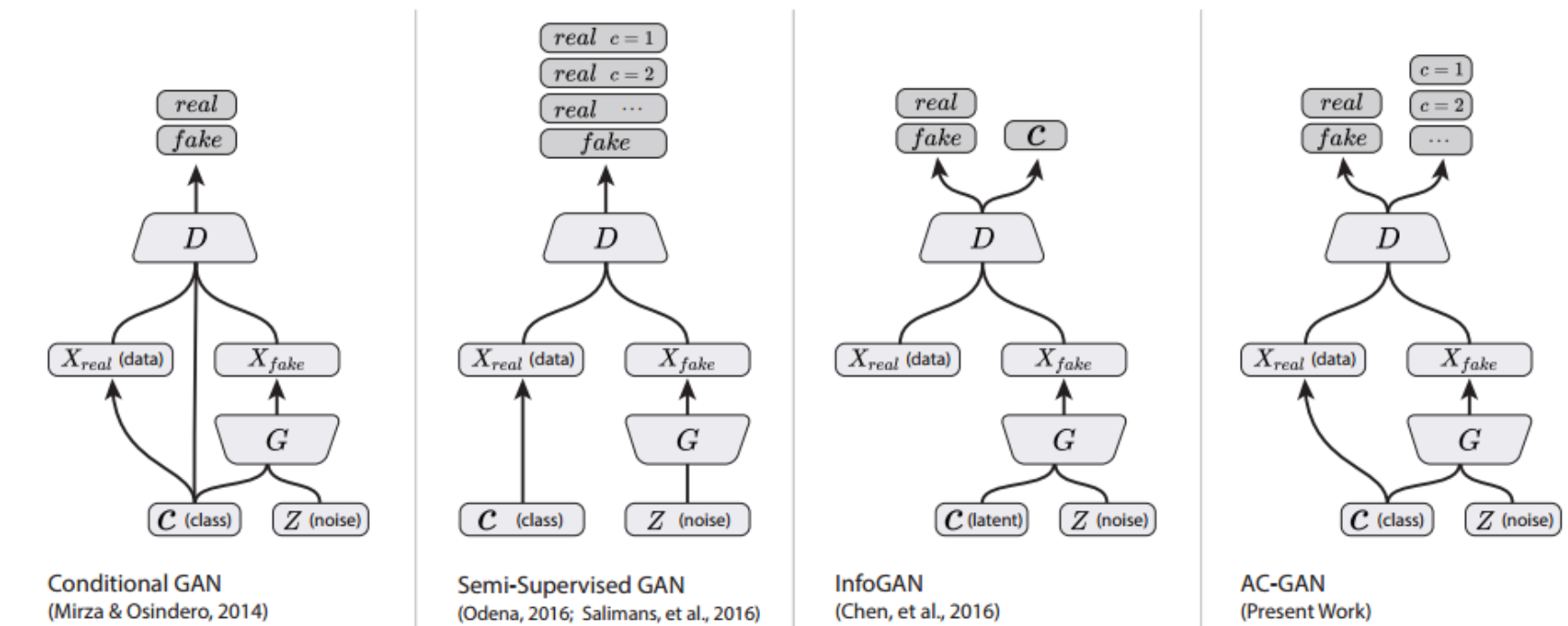


Figure: "Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks"

Name confused: deconvolution, transposed convolution, Fractionally Strided Convolution

Conditional adversarial

acGAN



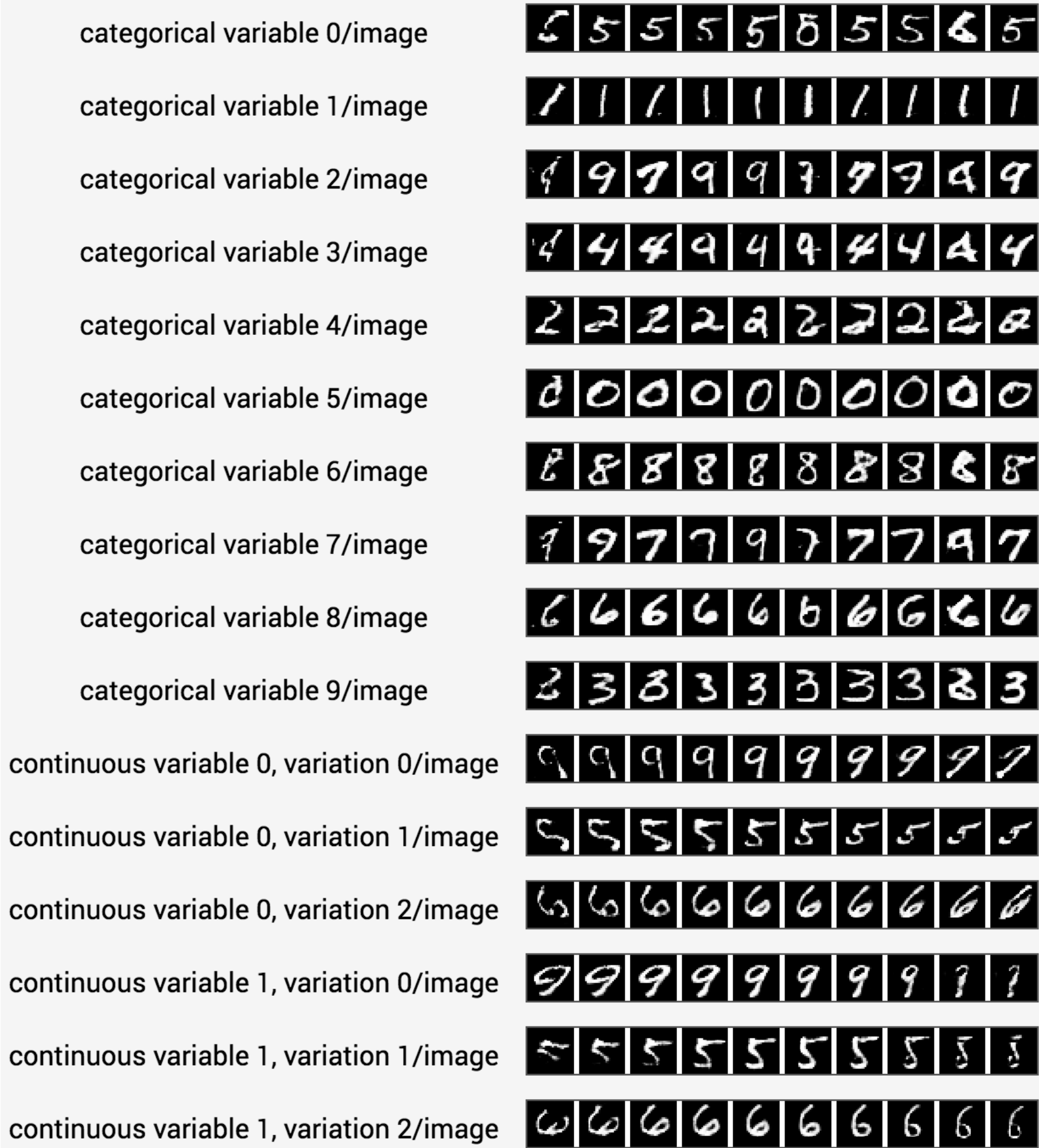
Paper: "Conditional Image Synthesis With Auxiliary Classifier GANs"

image to image



Paper: "Image-to-Image Translation with Conditional Adversarial Networks"

infoGAN



Invertible cGAN

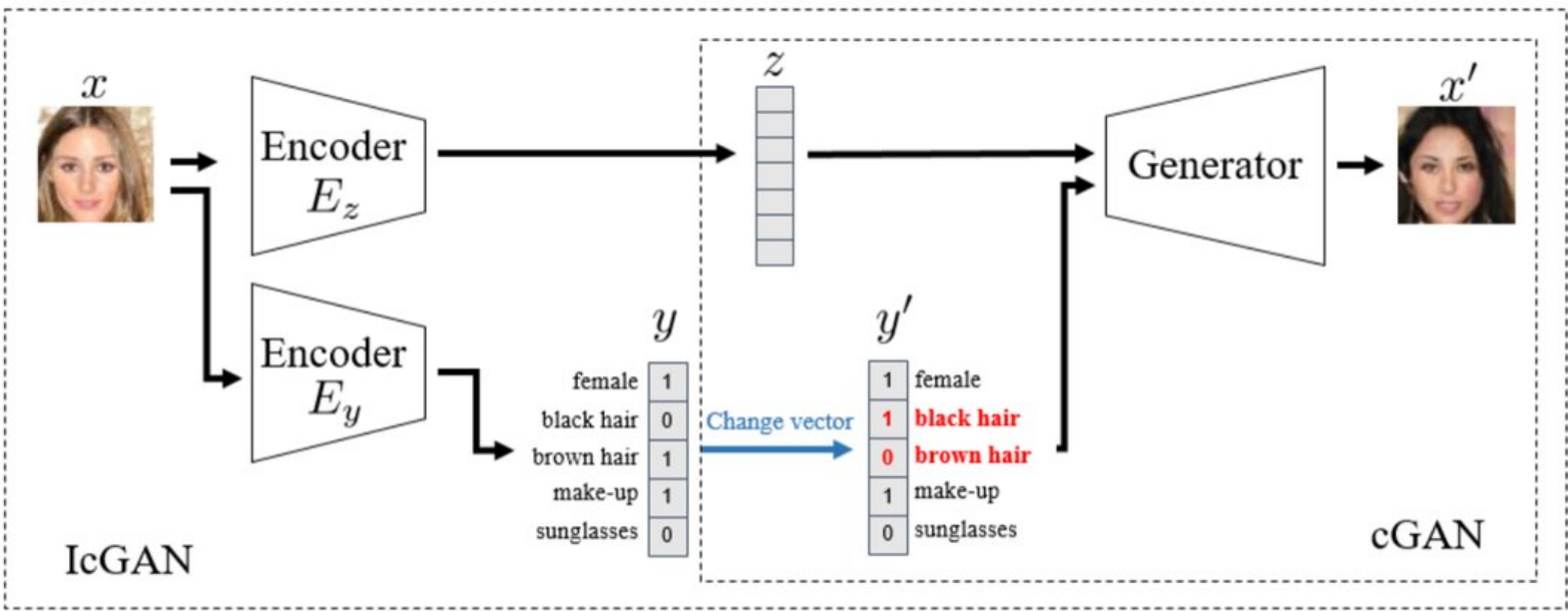
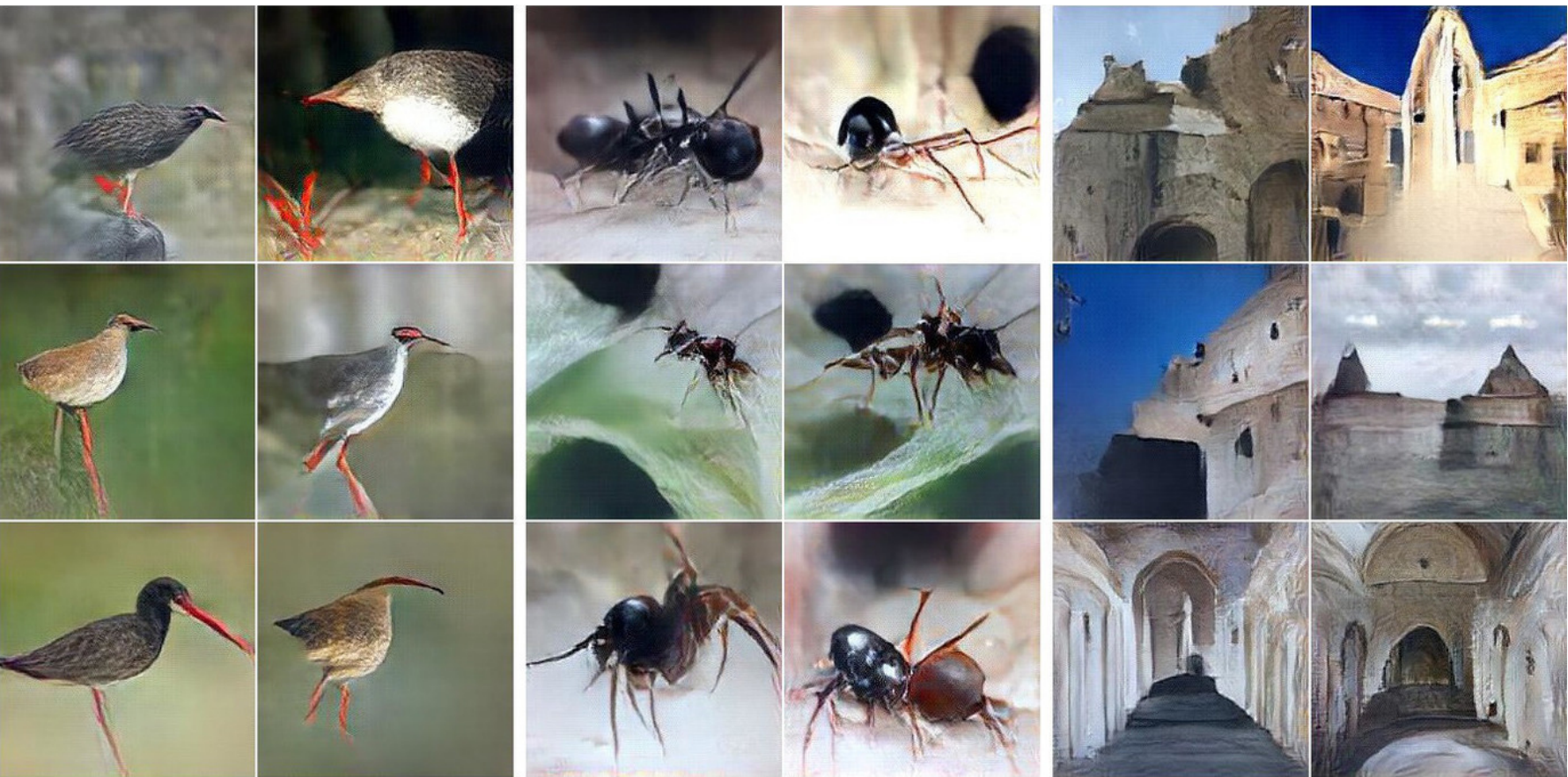


Figure 2: Scheme of a trained IcGAN, composed of an encoder (IND approach) and a cGAN generator. We encode a real image x into a latent representation z and attribute information y , and then apply variations on it to generate a new modified image x' .

State of the Art

PPGAN



redshank

ant

monastery



volcano

Paper: "Plug & Play Generative Networks: Conditional Iterative Generation of Images in Latent Space"

Theory Breakthrough

Wasserstein GAN

Contribution:

- Explain and solve training instability theoretically and practically.
- Don not need any fancy tricks (even MLP works).
- Have a better indicator to monitor the training process.
- More diverse images generated.

Solution:

- Use different loss function (without **log** function).
- Weights clip.

- Without optimization involving momentum.

Paper: "Towards Principled Methods for Training Generative Adversarial Networks"

Paper: "Wasserstein GAN"

Reference: [令人拍案叫绝的Wasserstein GAN](#)

Problems

- What is the right metric to quantify images' quality?
- Generator hard to generate discrete data (e.g. NLP).
- How can GAN be combined with our current works?

All in one page

[zhangqianhui/AdversarialNetsPapers](#)

The classical papers and codes about generative adversarial nets